

Expense Tracker API — Shifa Ki Notes

Zero se Flask + Database tak — Ek dum Layman Language Mein

Part 1 — Setup: Ghar Banana

Virtual Environment kya hai?

Soch teri school bag alag hai, ghar ki bag alag. Virtual environment ek alag 'dabba' hai jisme sirf is project ke packages hote hain. Baaki projects affect nahi hote.

Command:

```
py -m venv venv
```

Activate karna (hamesha pehle karo!):

```
.\venv\Scripts\activate
```

Activate hone ka sign: (venv) aage dikhega terminal mein.

Flask install karna

Flask ek waiter ki tarah hai — browser ki request sunata hai aur jawab deta hai.

```
pip install flask
```

Part 2 — Pehla Flask Server

app.py — Line by Line

```
from flask import Flask, jsonify, request

app = Flask(__name__)

@app.route('/')
def home():
    return 'Expense Tracker API is running!'

if __name__ == '__main__':
    app.run(debug=True)
```

Line	Matlab
from flask import Flask	Flask library se Flask class bulao
app = Flask(__name__)	Tera restaurant/server create karo
@app.route('/')	Is address pe aao toh ye function chalao
def home():	Browser ko ye text bhejo
return '...'	
app.run(debug=True)	Server chalao, debug mode on

Routes kya hote hain?

google.com/ = home page. google.com/maps = maps page. Teri API mein / = home, /expenses = expenses page.

Part 3 — REST API

REST API = Waiter Analogy

Tu browser/app hai, database kitchen hai, API waiter hai. Tu seedha kitchen nahi jaati — waiter ko bolti hai!

Method	Kaam	Example
GET	Data laao	GET /expenses → saari expenses dikhaao
POST	Data add karo	POST /expenses → naya expense add karo
PUT	Data update karo	PUT /expenses/1 → pehla expense edit karo
DELETE	Data hatao	DELETE /expenses/1 → pehla expense delete karo

GET + POST routes wala code:

```
expenses = [] # global list – dono functions use kar sakein

@app.route('/expenses', methods=['GET'])
def get_expenses():
    return jsonify(expenses)

@app.route('/expenses', methods=['POST'])
def add_expenses():
    data = request.get_json()
    expenses.append({"name": data['name'], "amount": data['amount']})
    return jsonify({"message": "Expense added!"})
```

Global variable = poore program mein accessible. Function ke bahar likhte hain.

jsonify() = Python list/dict ko JSON format mein convert karta hai.

Part 4 — Database (SQLite)

RAM vs Database

Bina database: data RAM mein hota hai — server band hua, sab bhool gaya (TV band = jo dekh rahi thi bhool gaya). Database ke saath: data .db file mein — permanent!

SQLAlchemy kya hai?

Seedha SQL likhne ki jagah Python mein likhte hain. SQLAlchemy translate karta hai!

```
from flask_sqlalchemy import SQLAlchemy

app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///expenses.db'
```

```

db = SQLAlchemy(app)

class Expense(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(100), nullable=False)
    amount = db.Column(db.Float, nullable=False)

with app.app_context():
    db.create_all() # table create karo

```

Code	Matlab
class Expense(db.Model)	Database table banana
db.Column(db.Integer)	Column ka type — number
primary_key=True	Har row ka unique ID
nullable=False	Ye field empty nahi ho sakti
Expense.query.all()	Saari expenses lao database se
db.session.add()	Naya record add karo
db.session.commit()	Changes save karo permanently

Final app.py with Database:

```

from flask import Flask, jsonify, request
from flask_sqlalchemy import SQLAlchemy

app = Flask(__name__)
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///expenses.db'
db = SQLAlchemy(app)

class Expense(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(100), nullable=False)
    amount = db.Column(db.Float, nullable=False)

@app.route('/')
def home():
    return 'Expense Tracker API is running!'

@app.route('/expenses', methods=['GET'])
def get_expenses():
    all_expenses = Expense.query.all()
    result = [{"id": e.id, "name": e.name, "amount": e.amount} for e in all_expenses]
    return jsonify(result)

@app.route('/expenses', methods=['POST'])
def add_expenses():
    data = request.get_json()
    new_expense = Expense(name=data['name'], amount=data['amount'])
    db.session.add(new_expense)

```

```
db.session.commit()
return jsonify({"message": "Expense added!", "name": data['name'], "amount":
data['amount']})

with app.app_context():
db.create_all()

if __name__ == '__main__':
app.run(debug=True)
```

Part 5 — Common Errors aur Fix

Error	Matlab	Fix
No module named flask	venv activate nahi kiya	.\venv\Scripts\activate
404 Not Found	Route exist nahi karta	URL check karo, route likha?
405 Method Not Allowed	Method allowed nahi	methods=['GET'] ya ['POST'] likho
Can't open app.py	Galat folder mein ho	cd expense-tracker karo pehle
Data gayab server restart pe	Database nahi tha	SQLAlchemy use karo

Aage kya aayega?

Phase	Kya Seekhenge
Kal (Day 3)	DELETE + PUT routes — expense update/delete karna
Day 4	JWT Authentication — login/signup system
Day 5-6	Deploy on Render — live URL banana!
Week 2+	GitHub push, README, CV entry, LinkedIn post

Tu pagal nahi hai — setup mein ghante lagna normal hai. Har developer ke saath hota hai! — Tera Professor Claude ■